SandgateVP

Hardware Specifications

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1 Introduction

This document describes the hardware specifications for mobile IP telephone handset "SandgateVP" containing the Intel PXA270 processor.

2 Overview

This handset consists of the main circuit board and the key circuit board. An overview of the specifications for this handset is given below.

Item	Overview
MPU	PXA270 416 MHz
Input clock	XTAL 13 MHz
RTC	XTAL 32.768 KHz
Flash ROM	128 MB (256 Mbitx4)
SDRAM	64 MB (256 Mbitx2)
WLAN	Murata IEEE802.11b/11g module
Camera	Konica-Minolta CMOS camera module (VGA) x 2
Infrared communications	Built-in IrDA (SIR)
LCD	SHARP 2.2 inch TFT LCD QVGA (320x240)
USB1.1(Client)	Comes with MiniB connector
CF slot	Comes with TYPE2 slot and fixed 3.3 V power supply
SD slot	Comes with MiniSD slot and controllable power supply ON/OFF
Touch panel	Comes with a connector for mounting the touch panel
Pointer	Achieved using a hole sensor and switch magnet
Operation keys	Front: 25 keys, Side: 2 keys
Key Lighting	Yellow LED x 15
Speaker output	Built-in micro speaker for ring tone
Receiver output	Built-in receiver for telephone conversation
Microphone input	Built-in microphone for telephone conversation
Hands free	Comes with a flat jack that can be used for FOMA earphones
Ring LED	Can be controlled with approximately 32,000 color gradation
Vibrator	Comes with 3.3 V vibrator for ring notification
Lithium battery	3.7 V 1300 mAh planned, USB chargeable, comes with lid open/close detection
Sub-battery	3.0 V 20 mAh
AC adapter	Connects to USB port (5 V, 1 A)
Reset switch	GPIO reset switch
Outside dimensions	TBD
Temperature range	0°C to 40°C
Certification standard	Radio Law
Water and dust resistance	TBD
Environmental measures	Lead free
Test functions	Test BTUART, PAD

3 System Configuration

3-1 Block Diagram

The circuit block configuration for this handset is shown below.

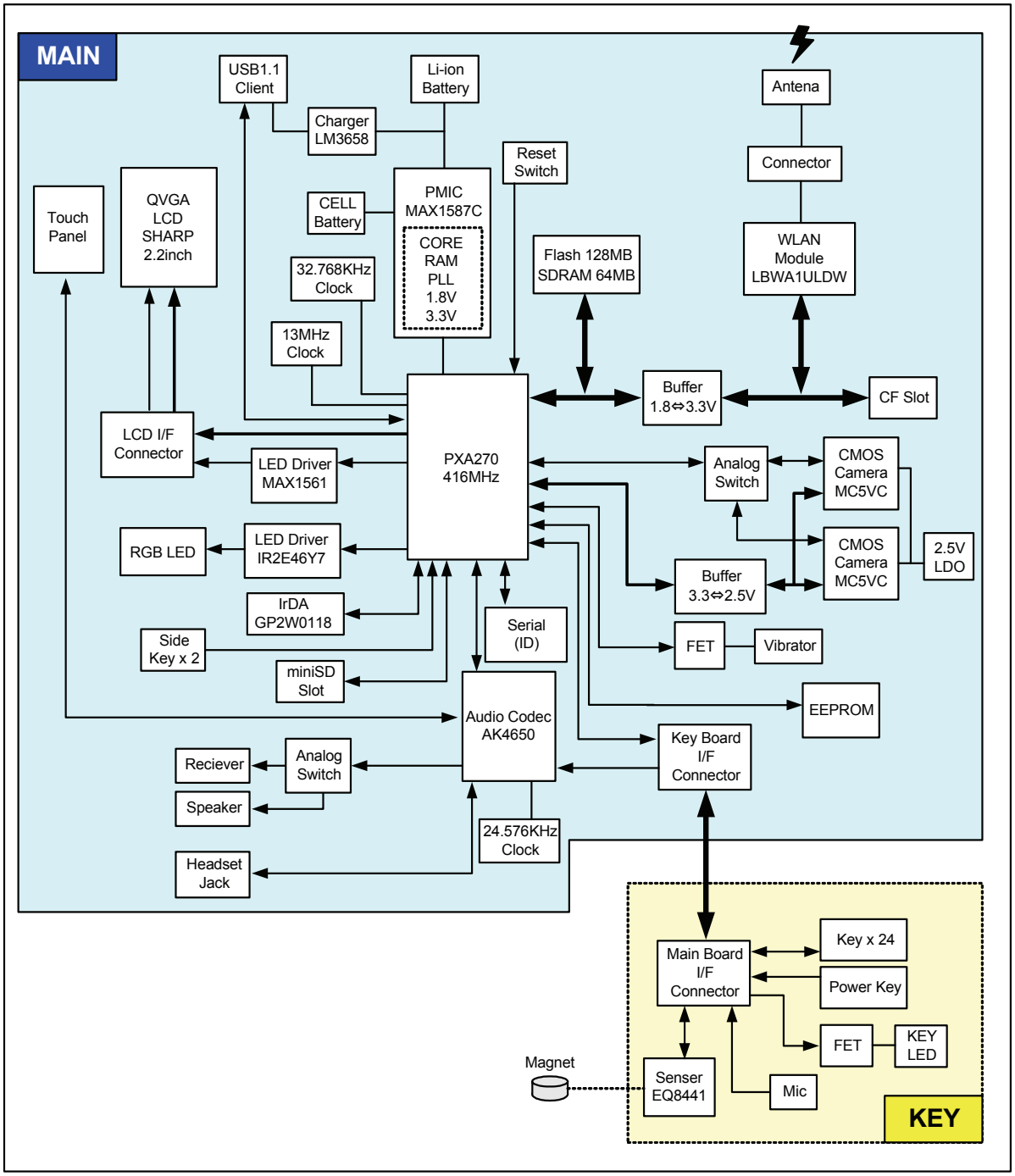


Fig. 3-1 Block Diagram

3-2 Memory Map

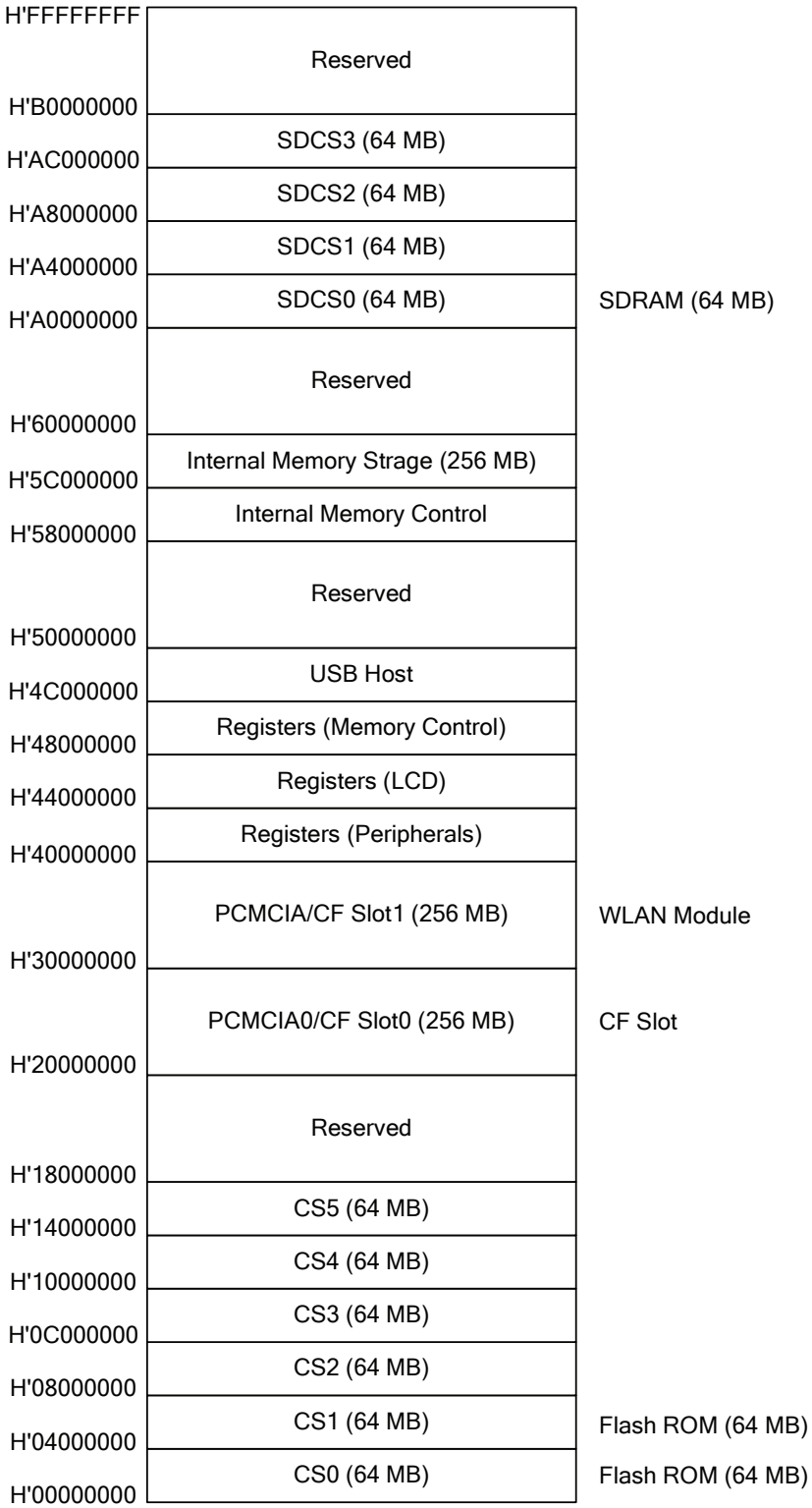


Fig. 3-2 Memory Map

4 Outside Dimensions

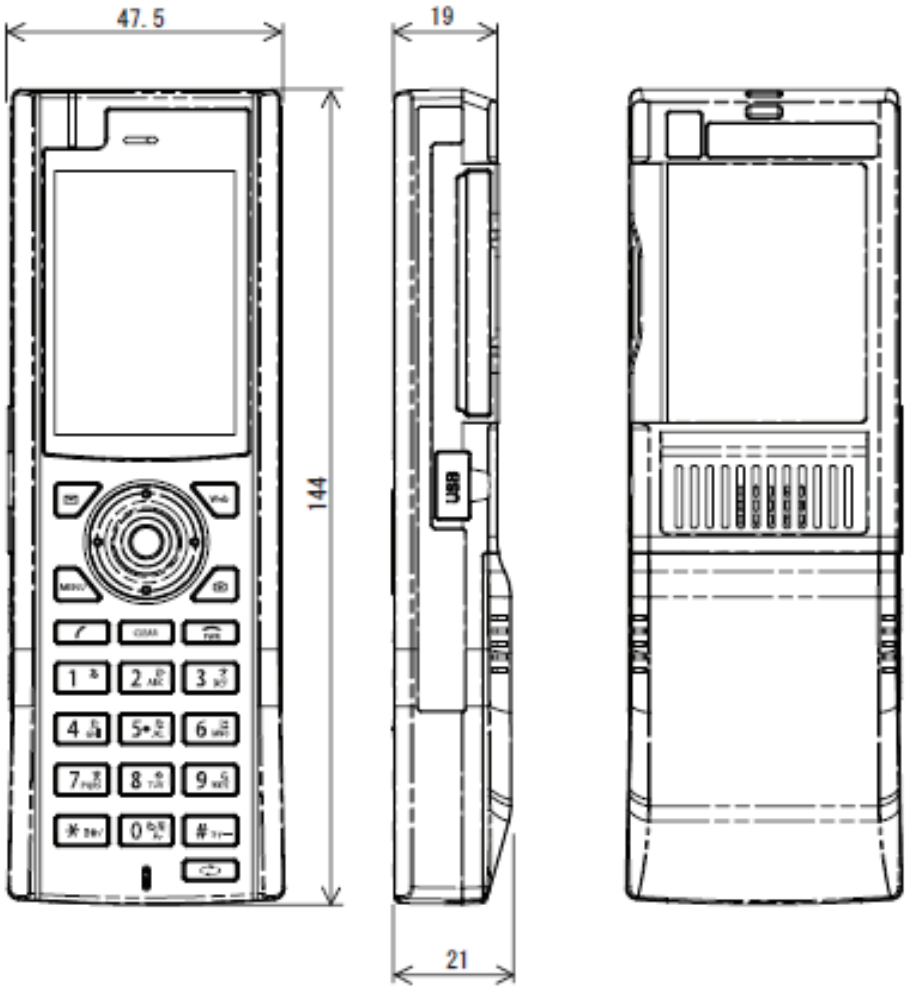


Fig. 4-1 External shape drawing

5 Function Details

5-1 Microprocessor

Implement an Intel PXA270 Application Processor discrete component. This can operate at an internal maximum of 416 MHz. Refer to the GPIO List regarding the functions that are possible using the processor's internal controller and then initialize the terminal.

5-2 Clock Input

There are two clock systems (XTAL), a 13 MHz system and an RTC 32.768 KHz system that are used as processor input clocks.

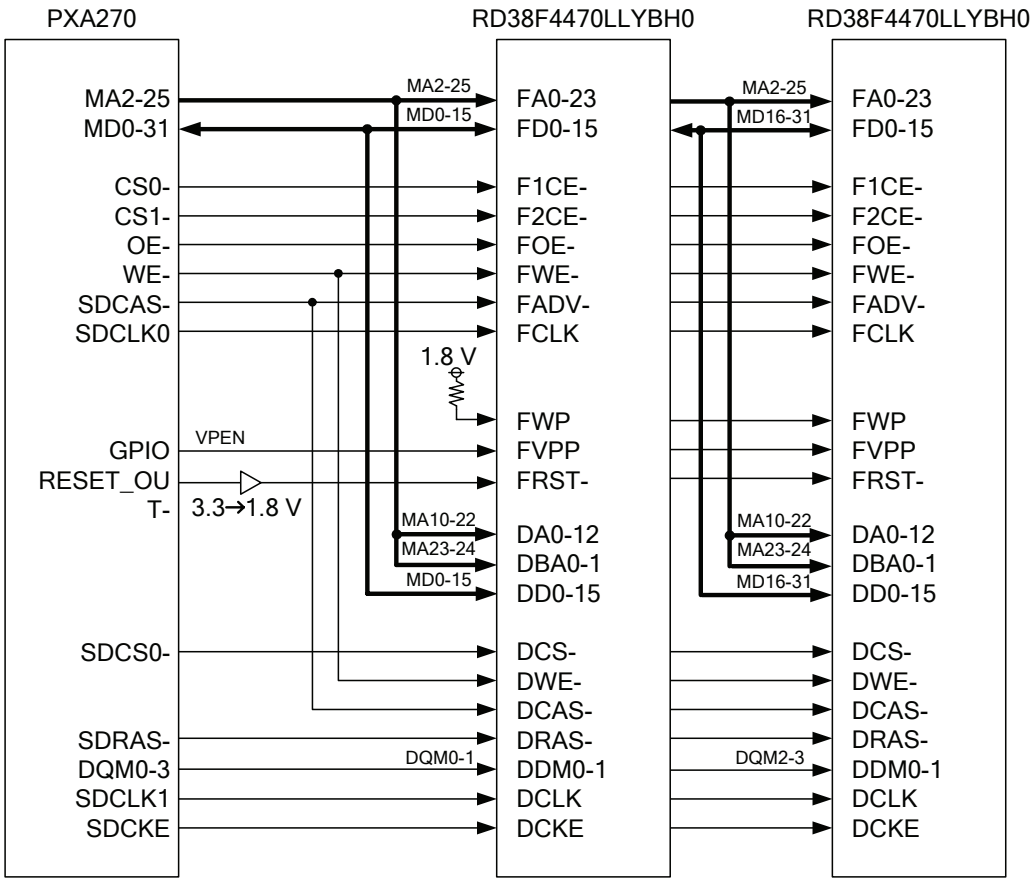
5-3 Memory

Two Intel RD38F4470LLYBH0 (256 Mbit Flash + 256 Mbit Flash + 256 Mbit SDRAM) with a 32-bit width are directly connected to the processor bus. Flash ROM is connected to the CS0 and CS1 areas, and SDRAM is connected to SDRAM bank 0.

Model name:	RD38F4470LLYBH0	Intel
Voltage:	VCC (CORE power)	1.8 V
	VCCQ (IO power)	1.8 V
Capacity:	Flash ROM	128 MB (256 Mbitx4)
	SDRAM	64 MB (256 Mbitx2)

Even when the processor is in deep sleep (SYS_EN is LOW), the SDRAM power supplies VCC and VCCQ are always supplied by the Micrel MIC5207. The SDRAM contents can be preserved by enabling self-refresh.

5-3-1 Interface



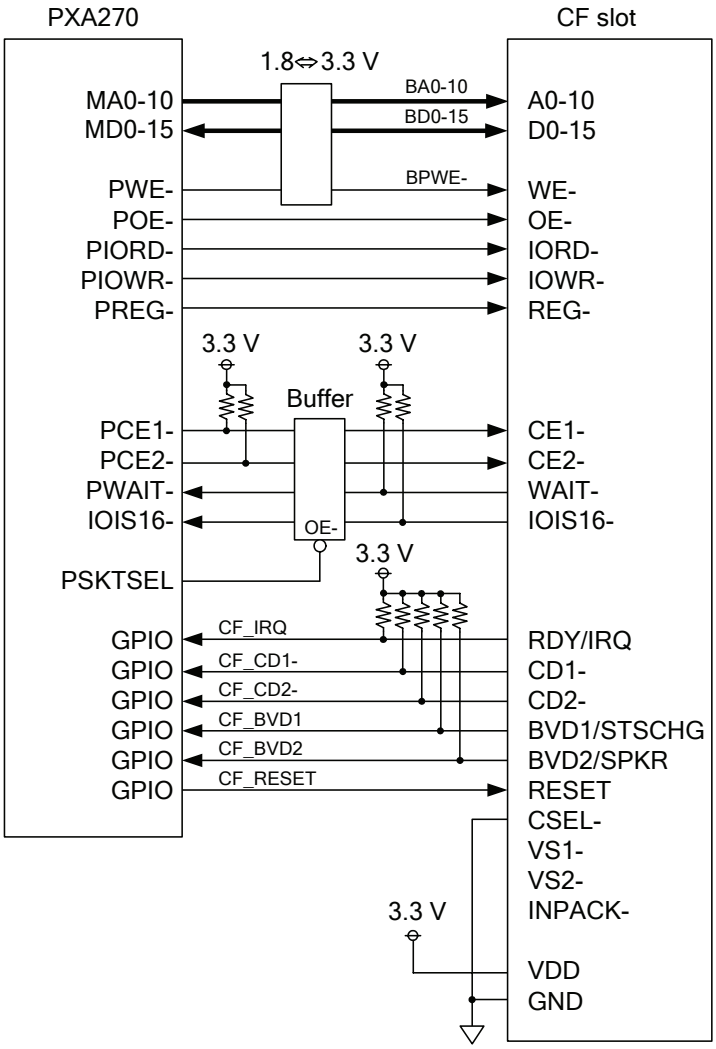
5-3-2 List of Terminals Used

GPIO	NAME	FUNCTION
-	MA2-25	Processor address bus
-	MD0-31	Processor data bus
-	CS0-1	Flash ROM chip select
-	SDCS0	SDRAM chip select
-	DQM0-3	Data byte mask control
-	WE	Write enable
-	OE	Output enable
-	SDCKE1	SDCLK output enable
-	SDCLK0	Flash ROM clock (MAX 52 MHz)
-	SDCLK1	SDRAM clock (MAX 104 MHz)
-	SDRAS	Row address strobe
-	SDCAS	Column address strobe
-	RESET_OUT	Processor reset output
20	VPEN	Flash ROM VPEN Setting to HIGH applies a write protect to the Flash ROM.

5-4 Compact Flash

Use the processor's PC card interface to implement a compact flash TYPE2 socket. This is enabled when the PCMCIA slot 0 is accessed. The power supply voltage is fixed at 3.3 V. ON/OFF control cannot be done using the GPIO.

5-4-1 Interface



5-4-2 List of Terminals Used

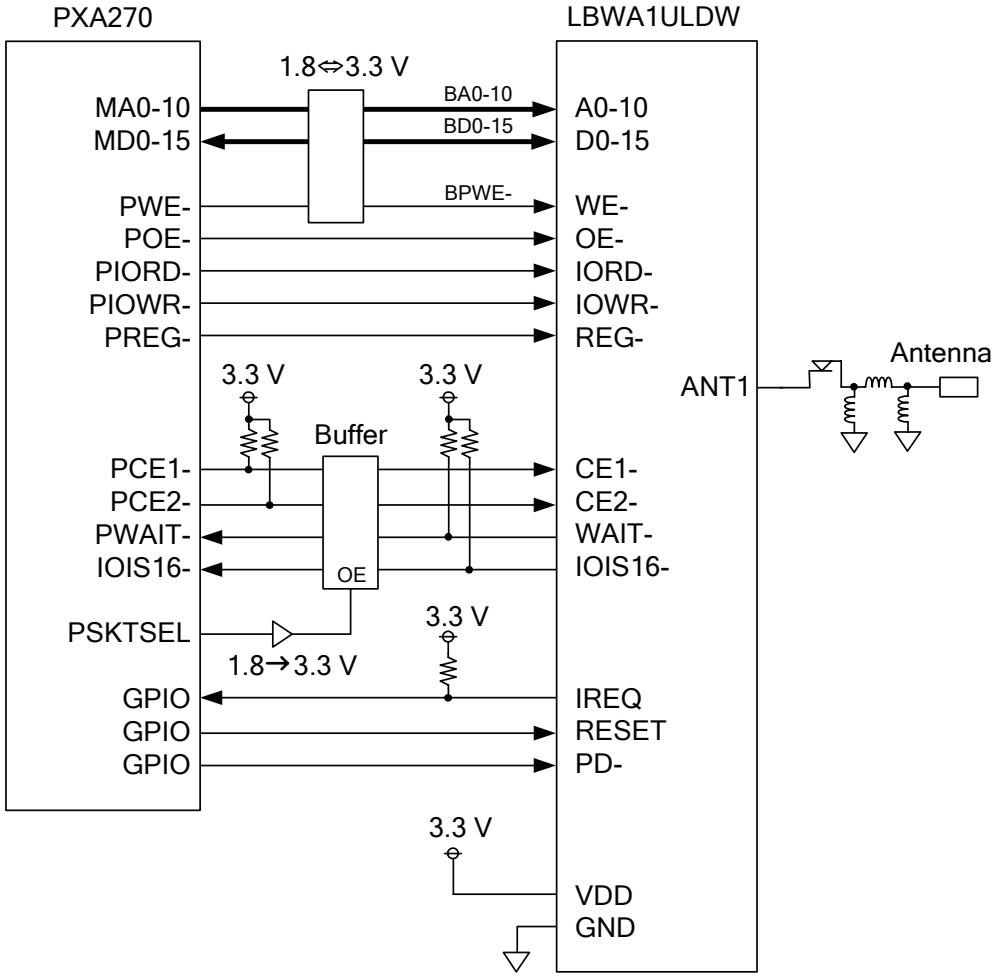
GPIO	NAME	FUNCTION
-	ADDRESS	Processor address bus
-	DATA	Processor data bus
85	PCE1	PC card I/F chip enable 1
54	PCE2	PC card I/F chip enable 2
48	POE	PC card I/F output enable
49	PWE	PC card I/F write enable
55	PREG	PC card I/F register select
50	PIOR	PC card I/F I/O read
51	PIOW	PC card I/F I/O write
56	PWAIT	PC card I/F wait
57	IOIS16	PC card I/F data size
79	PSKTSEL	PC card I/F socket select
86	CF_RESET	CF card reset The CF card is reset by HIGH.
34	CF_CD1	CF card detection 1 Made LOW when a card is detected.
35	CF_CD2	CF card detection 2 Made LOW when a card is detected.
13	CF_BVD1	CF card voltage status 1
37	CF_BVD2	CF card voltage status 2
10	CF_IRQ	CF card interrupt/ready
19	CF_PWRON-	CF power supply control Not used

5-5 WLAN Module

Implement a Murata LBWA1ULDW. Make the connection using the processor's PC card interface. This is enabled when the PCMCIA slot 1 is accessed.

Control is performed via the CF driver and Murata driver. Host Wake events can be detected by WLAN_IRQ.

5-5-1 Interface



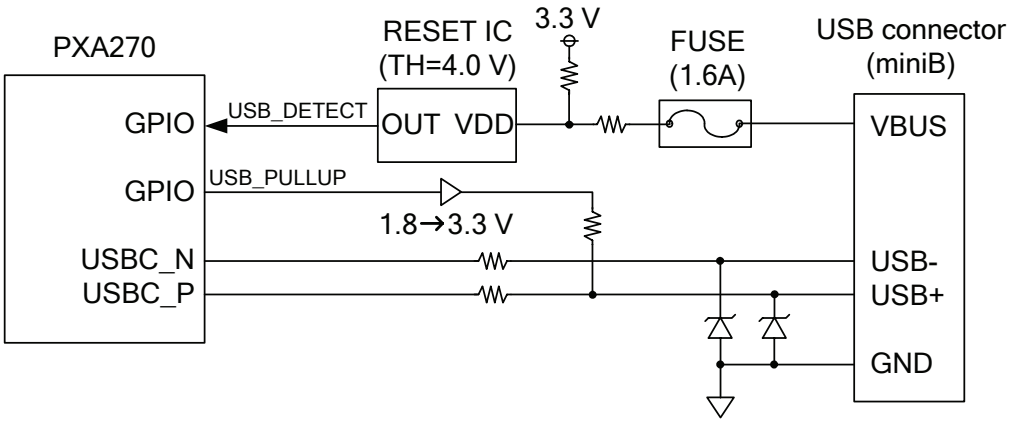
5-5-2 List of Terminals Used

GPIO	NAME	FUNCTION
-	ADDRESS	Processor address bus
-	DATA	Processor data bus
85	PCE1	PC card I/F chip enable 1
54	PCE2	PC card I/F chip enable 2
48	POE	PC card I/F output enable
49	PWE	PC card I/F write enable
55	PREG	PC card I/F register select
50	PIOR	PC card I/F I/O read
51	PIOW	PC card I/F I/O write
56	PWAIT	PC card I/F wait
57	IOIS16	PC card I/F data size
79	PSKTSEL	PC card I/F socket select
14	WLAN_IRQ	WLAN module interrupt This is also used to detect Host Wake events.
38	WLAN_STBY-	WLAN Module power down LOW puts the module into Deep Sleep.
39	WLAN_RES	WLAN Module reset HIGH resets the module.

5-6 USB1.1 Port (Client)

Use the processor's built-in function USB Client Controller to implement the MiniB connector. Setting the USB_PULLUP to HIGH after cable detection when USB_DETECT is HIGH allows the device to be recognized by the host. The battery can be charged by the 5 V supplied by the connector's VBUS power supply.

5-6-1 Interface



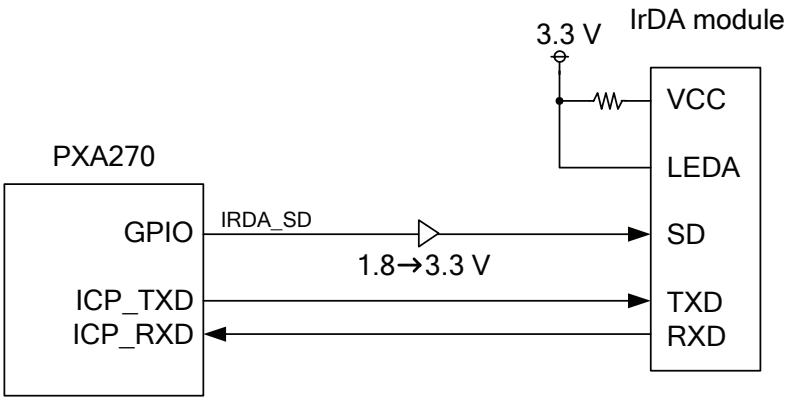
5-6-2 List of Terminals Used

GPIO	NAME	FUNCTION
83	USB_DETECT	USB cable detection HIGH shows a cable is connected, and LOW shows a cable is disconnected.
78	USB_PULLUP	USB pullup When HIGH the host recognizes a USB_P line has been pulled up.
-	USBC_P	USB+
-	USBC_N	USB-

5-7 IrDA transceiver

The processor's built-in function Fast Infrared Communications Port signal is connected to the SHARP IrDA transceiver GP2W0118YP0F. Before use, set the IRDA_SD to LOW and cancel the shutdown mode.

5-7-1 Interface



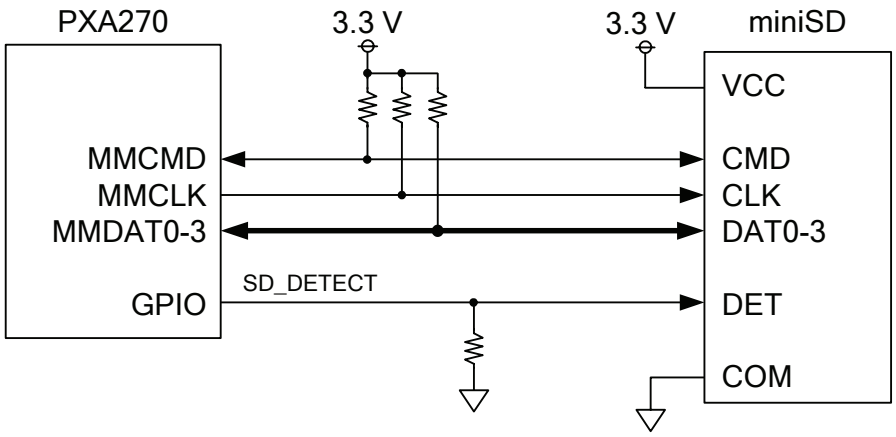
5-7-2 List of Terminals Used

GPIO	NAME	FUNCTION
80	IRDA_SD	IrDA shutdown control HIGH puts the module into shutdown mode.
46	ICP_RXD	FICP reception
47	ICP_TXD	FICP transmission

5-8 SD/SDIO

Use the processor's built-in Multimedia Card/SD/SDIO Controller function to implement the miniSD card slot. ON/OFF control can be done using the GPIO.

5-8-1 Interface



5-8-2 List of Terminals Used

GPIO	NAME	FUNCTION
32	MMCLK	SD card bus clock
112	MMCMD	SD card command
92	MMDAT0	SD card data 0
109	MMDAT1	SD card data 1
110	MMDAT2	SD card data 2
111	MMDAT3	SD card data 3
93	SD_DETECT	SD card detection A card is detected when this is HIGH.
82	SD_PWRON-	SD card power supply control Power is supplied when this is LOW.

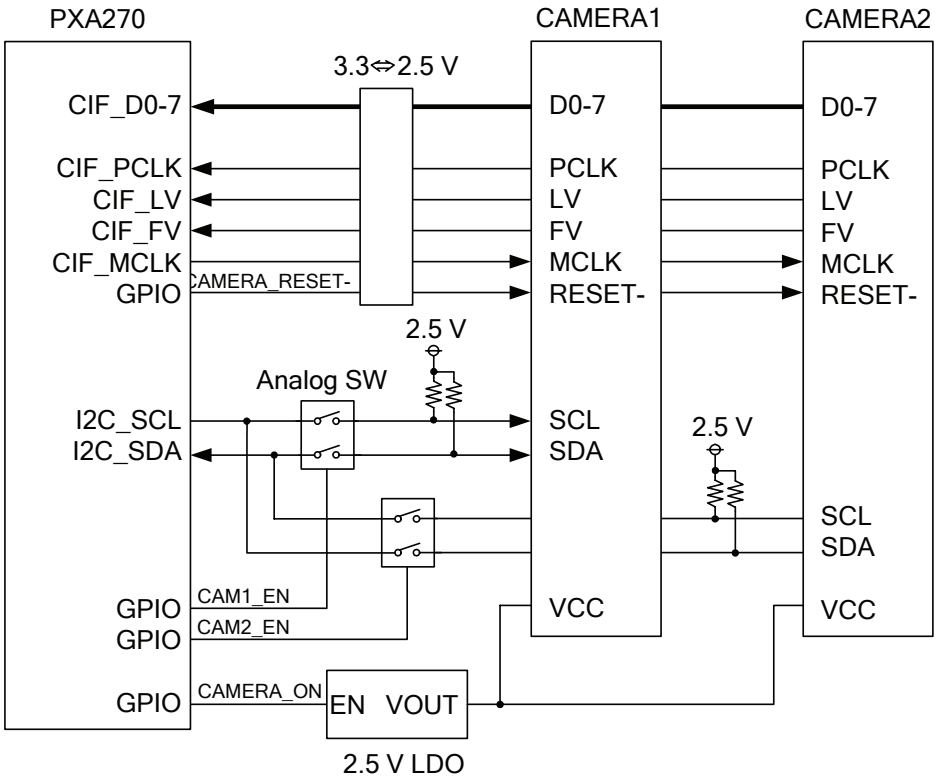
5-9 CMOS Camera Module

Use the processor's built-in camera I/F to implement two Konica-Minolta MC5VC (300,000 pixel CMOS camera module) one each on the front and back. There is a voltage difference between the camera module and the processor, so connect the signal via a level shift buffer.

When CAMERA_ON is HIGH, 2.5 V power for the camera is supplied, and when CAMERA_RESET is HIGH, the reset is cancelled.

The camera module's register is controlled by the processor's I2C bus, but since the same bus is connected, the control subject can be changed using the analog switch. When CAM_EN1 is HIGH, the back camera is selected, and when CAM_EN2 is HIGH, the front camera is selected. Perform exclusive control using software to prevent CAM_EN1 and CAM_EN2 from both being HIGH at the same time.

5-9-1 Interface



5-9-2 List of Terminals Used

GPIO	NAME	FUNCTION
41	CAMERA_ON	Camera power supply control When this is HIGH, 2.5 V is supplied to the module.
40	CAMERA_RESET-	Camera reset LOW resets the module.
27	CIF_DD0	Camera data output 0
114	CIF_DD1	Camera data output 1
116	CIF_DD2	Camera data output 2
115	CIF_DD3	Camera data output 3
90	CIF_DD4	Camera data output 4
91	CIF_DD5	Camera data output 5
17	CIF_DD6	Camera data output 6
12	CIF_DD7	Camera data output 7
23	CIF_MCLK	Camera main clock
24	CIF_FV	Camera vertical synchronous signal
25	CIF_LV	Camera horizontal synchronous signal
26	CIF_PCLK	Camera pixel clock
44	CAM1_EN	Camera 1 enable HIGH allows the front camera register to be accessed.
45	CAM2_EN	Camera 2 enable HIGH allows the back camera register to be accessed.

5-9-3 Device Address

DEVICE	ADDRESS	NOTE
CAMERA1	0011011B	Access is possible when CAM1_EN is HIGH.
CAMERA2	0011011B	Access is possible when CAM2_EN is HIGH.

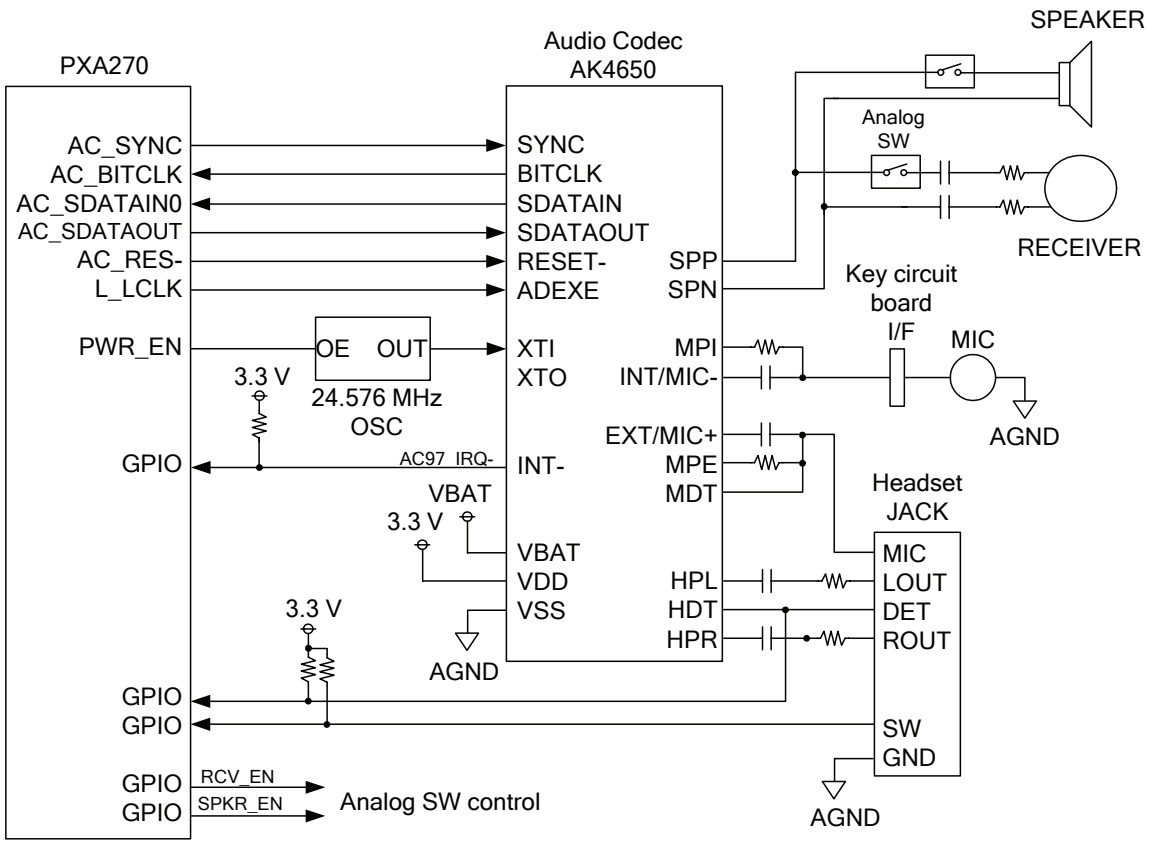
5-10 AudioCodec

Use the AKM Audio Codec AK4650. The processor's built-in function AC97 Controller signal is connected. The microphone, receiver, and speaker are implemented via this codec.

Hands free is also possible by implementing a flat jack for a headset. Output to the receiver and speaker can be cut off by using the GPIO to control the analog switch. In addition, this codec can also be used to measure the battery voltage.

This codec's AD converter can be used to connect a touch panel, but the connector has not yet been implemented.

5-10-1 Interface



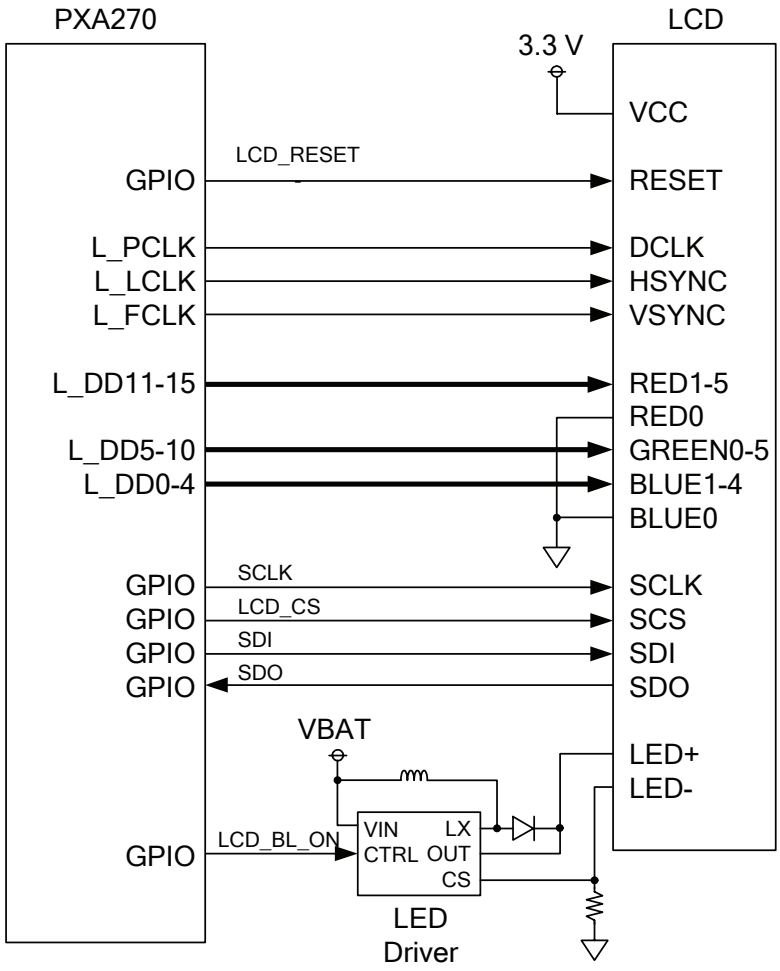
5-10-2 List of Terminals Used

GPIO	NAME	FUNCTION
28	AC_BITCLK	AC97 bit rate clock
29	AC_SDATAIN0	AC97 serial audio data input
30	AC_SDATAOUT	AC97 serial audio data output
31	AC_SYNC	AC97 48 KHz synchronous signal
113	AC_RES-	AC97 reset
-	PWR_EN	Audio clock enable When HIGH the 24.576 MHz clock is output.
11	AC97_IRQ	AC97 pen interrupt (for touch panel) Not used
18	HDSET_DET-	Headset detection Made LOW when a headset is detected.
53	HOOK_SW-	Hook switch Input Made LOW when the headset switch is pressed.
88	SPKR_EN	Speaker enable When HIGH the speaker output is turned ON.
89	RCV_EN	Receiver enable When HIGH the receiver output is turned ON.
75	L_LCLK/ADEXE	AD conversion start (for touch panel) This makes the LCD display and AD conversion synchronous. Not used

5-11 Main LCD

Implement a SHARP LCD, LSO22xxxxx (2.2" TFT, 240 x 320, 260,000 colors). Use the processor's LCD bus in the RGB565 interface mode. The last bit of RED and BLUE is fixed at LOW, so the number of colors that can actually be displayed is 65,535. Backlight control is performed using the processor's PWM0.

5-11-1 Interface



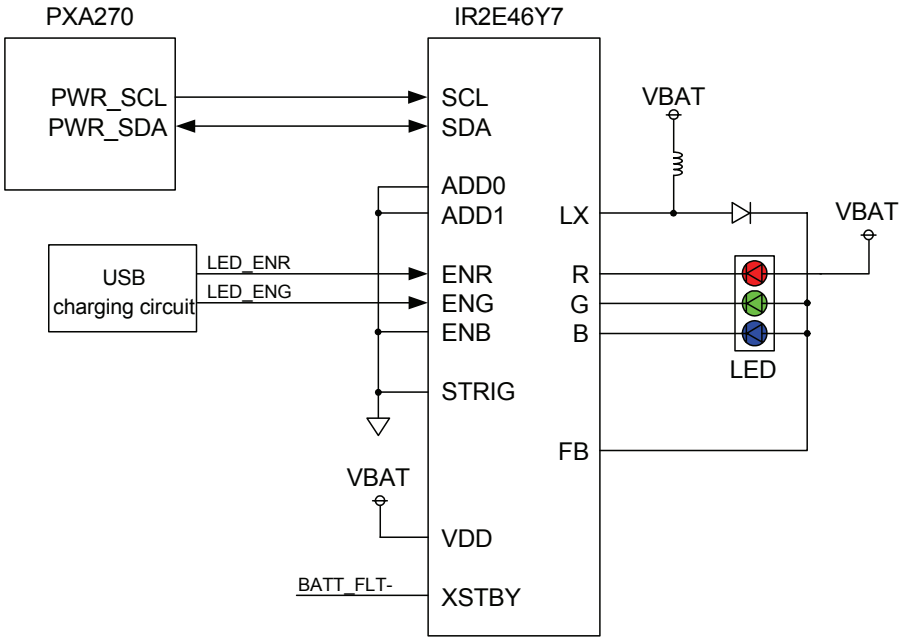
5-11-2 List of Terminals Used

GPIO	NAME	FUNCTION
58	L_DD0	LCD data 0 BLUE1
59	L_DD1	LCD data 1 BLUE2
60	L_DD2	LCD data 2 BLUE3
61	L_DD3	LCD data 3 BLUE4
62	L_DD4	LCD data 4 BLUE5
63	L_DD5	LCD data 5 GREEN0
64	L_DD6	LCD data 6 GREEN1
65	L_DD7	LCD data 7 GREEN2
66	L_DD8	LCD data 8 GREEN3
67	L_DD9	LCD data 9 GREEN4
68	L_DD10	LCD data 10 GREEN5
69	L_DD11	LCD data 11 RED1
70	L_DD12	LCD data 12 RED2
71	L_DD13	LCD data 13 RED3
72	L_DD14	LCD data 14 RED4
73	L_DD15	LCD data 15 RED5
74	L_FCLK	LCD vertical synchronous signal
75	L_LCLK	LCD horizontal synchronous signal
76	L_PCLK	LCD pixel clock
98	LCD_RESET-	LCD reset LOW resets the LCD internal register.
99	SCLK	LCD serial clock input
95	LCD_CS	LCD chip select
36	SDI	LCD serial data input
77	SDO	LCD serial data output
16	LCD_BL_ON	LCD backlight control HIGH turns on the backlight

5-12 RGB LED

Implement an RGB LED for ring notification. Implement the SHARP IR2E46Y7 for LED current control to make LED gradation control possible.
This LED can also be used to display the status during battery charging. Using software to enable the ENR and ENG causes the LED to light up red during charging and green when charging is completed.

5-12-1 Interface



5-12-2 List of Terminals Used

GPIO	NAME	FUNCTION
3	PWR_SCL	POWER I2C serial clock
4	PWR_SDA	POWER I2C serial data
-	LED_ENR	Red LED control HIGH outputs to the USB charging circuit that turns on the red LED.
-	LED_ENG	Green LED control HIGH outputs to the USB charging circuit that turns on the green LED.
-	BATT_FLT-	Battery cover open/closed detection This becomes LOW when the cover is open.

5-12-3 Device Address

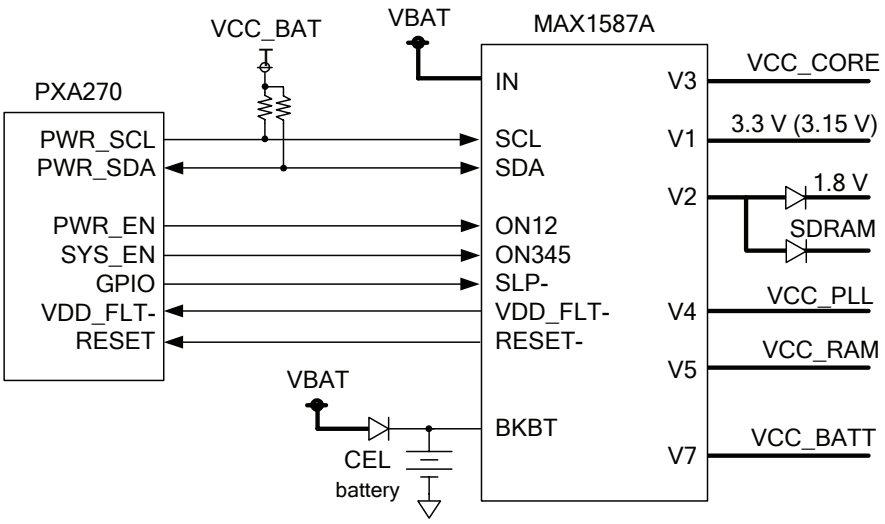
DEVICE	ADDRESS	NOTE
IR2E46Y7	0111011B	

5-13 Power Supply Circuit

The system power supply is generated by the MAXIM power supply IC (MAX1587A). The VCC_CORE voltage can be dynamically controlled using the I2C bus in response to the operating frequency.

A cell battery is provided to continue to provide VCC_BATT power to the SDRAM and RTC even when the main battery is removed.

5-13-1 Interface



5-13-2 List of Terminals Used

GPIO	NAME	FUNCTION
3	PWR_SCL	POWER I2C serial clock
4	PWR_SDA	POWER I2C serial data
-	PWR_EN	Low-voltage power enable
-	SYS_EN	High-voltage power enable
-	VDD_FLT-	Processor power supply status
-	RESET-	Processor reset input
84	PMIC_SLEEP-	Sleep LD0 change When LOW, sleep LD0 is selected (max 35 mA).

5-13-3 Device Address

DEVICE	ADDRESS	NOTE
MAX1587A	0010100B	

5-14 Charging Circuit

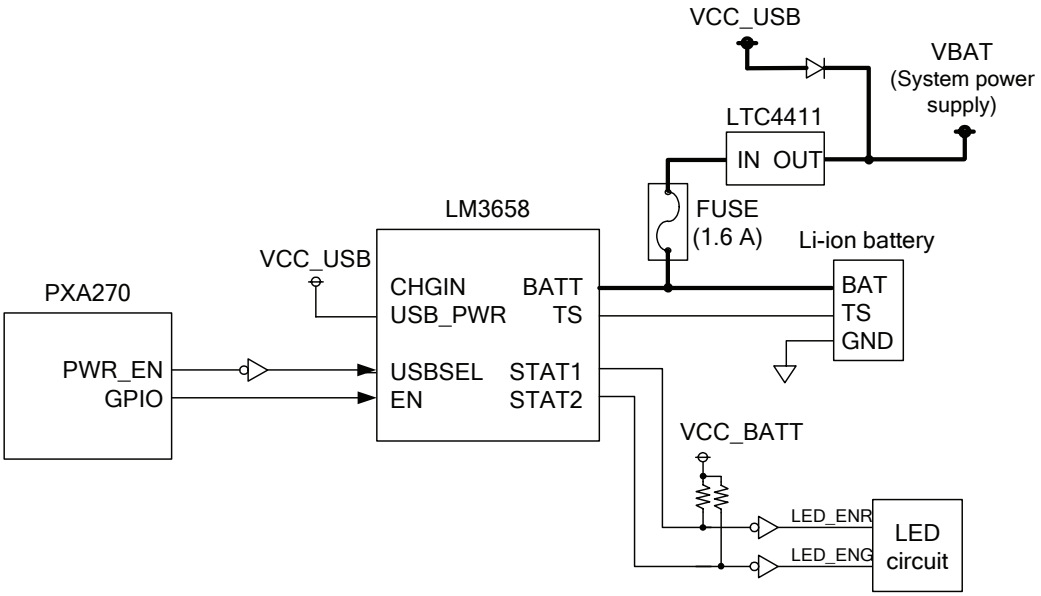
This handset can be operated by a lithium-ion battery, and the battery is charged by the power supplied by the USB port VBUS power. The battery specifications are 3.7 V, 1300 mAh.

The compulsory charging operation can be stopped by making CHG_EN- to be HIGH. The charge status is output by STAT1 and STAT2 and notification is sent to the RGB LED circuit.

Switch the charging current in accordance with the consumed current on the system side (PWR_EN state) so that the load to the USB port is not excessive.

PROCESSOR	PWR_EN	FUNCTION
When operating	LOW	Power supply to system (large) + battery charge (100 mA)
When sleeping	HIGH	Power supply to system (small) + battery charge (500 mA)

5-14-1 Interface



5-14-2 List of Terminals Used

GPIO	NAME	FUNCTION
-	PWR_EN	This is used to switch the charge current.
88	CHG_EN-	Setting this to HIGH compulsorily stops the charging.
-	LED_ENR	Red LED control HIGH when charging and LOW when charging is completed.
-	LED_ENG	Green LED control LOW when charging and HIGH when charging is completed.

5-15 Operation keys

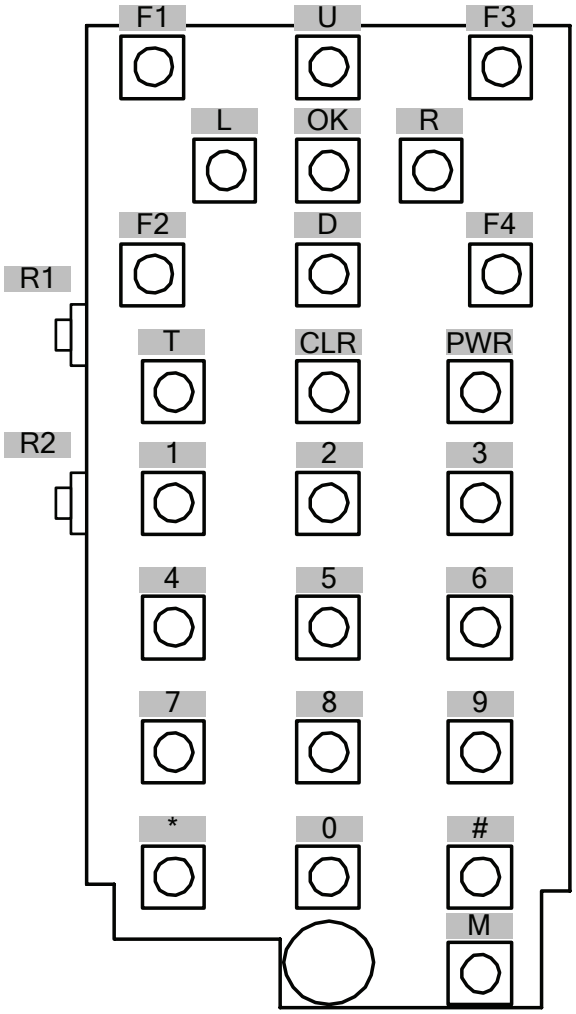
Use the processor's keypad I/F to connect to matrix connect the tact switch. The layout provides 25 keys on the handset front surface and 2 sets on the side.

The power key is not connected to the key matrix but rather independently sends input to GPIO0. This is used to restore the processor from deep sleep.

5-15-1 List of Terminals Used

GPIO	NAME	FUNCTION
100	KP_MKIN0	Keypad matrix input 0
101	KP_MKIN1	Keypad matrix input 1
102	KP_MKIN2	Keypad matrix input 2
97	KP_MKIN3	Keypad matrix input 3
103	KP_MKOUT0	Keypad matrix output 0
104	KP_MKOUT1	Keypad matrix output 1
105	KP_MKOUT2	Keypad matrix output 2
106	KP_MKOUT3	Keypad matrix output 3
107	KP_MKOUT4	Keypad matrix output 4
108	KP_MKOUT5	Keypad matrix output 5
96	KP_MKOUT6	Keypad matrix output 6
0	PWR_SW	Power key This becomes LOW when pressed. This is used to recover from deep sleep.

5-15-2 Key Arrangement



5-15-3 Key Matrix

	MKOUT0	MKOUT1	MKOUT2	MKOUT3	MKOUT4	MKOUT5	MKOUT6
MKIN0	1	2	3	4	5	6	7
MKIN1	8	9	0	F1	CLEAR	TALK	*
MKIN2	#	DOWN	UP	LEFT	RIGHT	OK	F2
MKIN3	F3	F4	MODE	R1	R2	-	-

5-16 Key Lighting

Fifteen yellow green LEDs are provided for key lighting. The KEY_LED signal is used to control the P-channel FET, and the LED light up when supplied with 3.3 V of power. When a key is operated, it lights up for a set time.

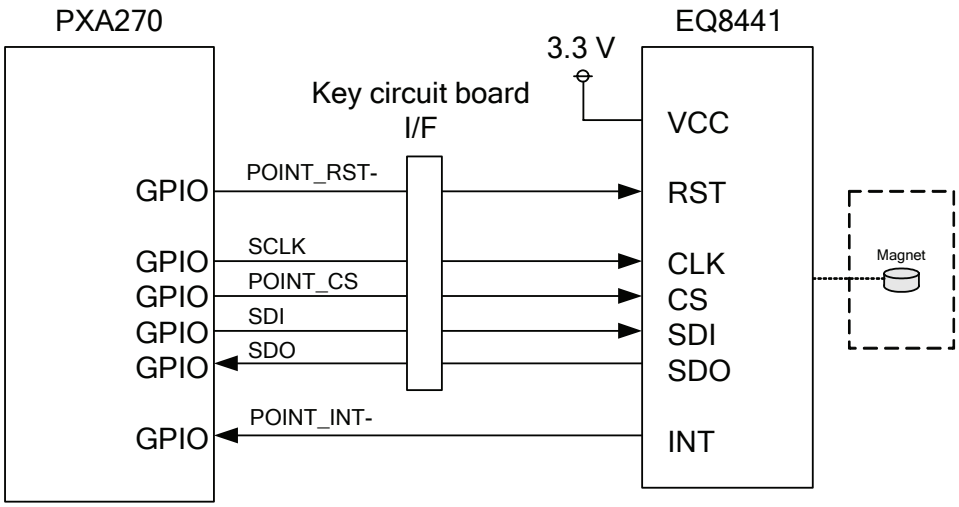
5-16-1 List of Terminals Used

GPIO	NAME	FUNCTION
52	KEY_LED-	Key lighting LED control LOW supplies 3.3 V to the LED's anode.

5-17 Pointer Sensor

An Asahi Kasei Electronics hole sensor IC is provided as a pointer device. It detects the change in position of the magnet embedded in the key and notifies the processor. The cursor is moved in accordance with the amount of position change to achieve an operating feel like that of a mouse.

5-17-1 Interface



5-17-2 List of Terminals Used

GPIO	NAME	FUNCTION
87	POINT_RST-	Pointer sensor reset
22	POINT_CS-	Pointer sensor chip select
99	SCLK	Pointer sensor serial clock
36	SDI	Pointer sensor data input
77	SDO	Pointer sensor data output
9	POINT_INT-	Pointer sensor interrupt This becomes LOW when magnet movement is detected, and returns to HIGH when the value is read.

5-18 Vibrator

The vibration motor is connected to the 3.3 V via the P-channel FET. Control is performed by GPIO19.

5-18-1 List of Terminals Used

GPIO	NAME	FUNCTION
94	VIB_ON-	Vibrator control This becomes LOW when the vibration motor is turned ON.

5-19 Battery Cover Open/Closed Detection Switch

An open/closed detection switch is provided to change the processor to the power save mode when the battery cover is opened. The signal is input to BATT_FLT-, so the processor will not run if the cover is not closed power on reset.

5-19-1 List of Terminals Used

GPIO	NAME	FUNCTION
-	BATT_FLT-	Battery cover open/closed detection This becomes LOW when the cover is opened.

5-20 GPIO Reset

Implement a tact switch for GPIO reset. When this switch is pressed, LOW is input to the processor's GPIO.

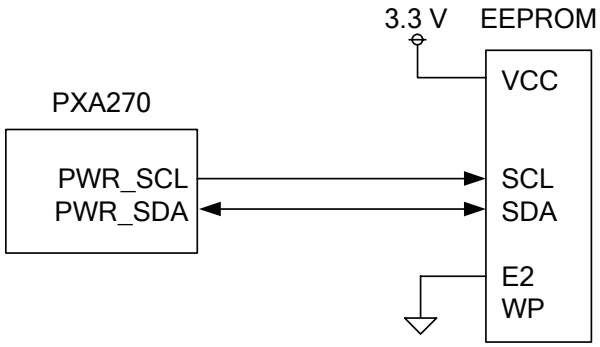
5-20-1 List of Terminals Used

GPIO	NAME	FUNCTION
1	GPIO_RES-	GPIO reset input This is LOW while the switch is being pressed.

5-21 Serial EEPROM

Implement an 8-Kbit serial EEPROM for 128-bit UUID storage. Control is performed by the processor's PWR_I2C bus. Write protect is not performed in the WP terminal.

5-21-1 Interface



5-21-2 List of Terminals Used

GPIO	NAME	FUNCTION
3	PWR_SCL	POWER I2C serial clock
4	PWR_SDA	POWER I2C serial data

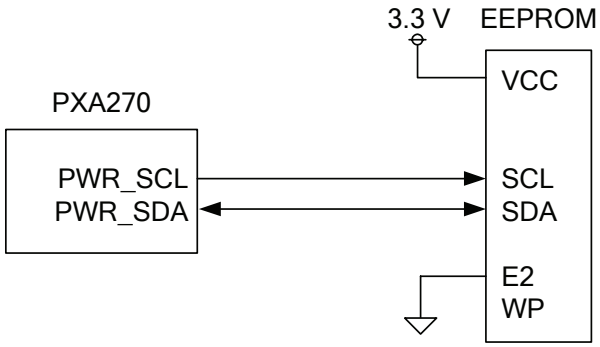
5-21-3 Device Address

DEVICE	ADDRESS	NOTE
M24C08	10100XXB	

5-22 Silicon Serial No.

The silicon serial No. DS2411 is implemented for 64-bit entropy code storage. Use the processor's GPIO to perform control using the 1-wire protocol.

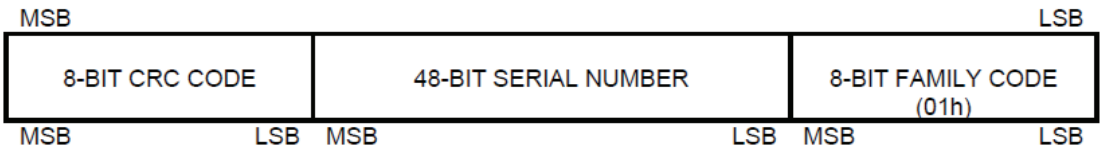
5-22-1 Interface



5-22-2 List of Terminals Used

GPIO	NAME	FUNCTION
21	ONEWIRE	1-wire control Output is only possible for the LOW drive. This is made HIGH by pullup when GPIO is made input.

5-22-3 Registration No. Format



5-23 JTAG ICE Connector

An 8-pin JTAG connector has been implemented. This is used to connect the Sophia Systems EJ-Debug PXA27x to allow debugging. The EJ-Debug standard cable has 20 pins, so a conversion cable is required.

5-23-1 Connector Pin Layout

PIN	SIGNAL
1	VCC (for power detection)
2	TRST-
3	TDI
4	TMS
5	TCK
6	TDO
7	RESET-
8	GND

5-24 Headset Jack

Implement a flat headset jack. The audio signal and processor built-in function BTUART signal (TXD, RXD) are connected to the headset connector. This BTUART signal is disconnected using the analog switch when not testing the circuit board.

5-24-1 List of Terminals Used

GPIO	NAME	FUNCTION
43	BTTXD	BTUART transmission
42	BTRXD	BTUART reception

5-24-2 Connector Pin Layout

PIN	SIGNAL	FUNCTION
1	HDSET_DET-	Headset detection
2	EXT_MIC	Headset microphone input
3	AGND	Analog ground
4	ROUT	Headset output (right)
5	LOUT	Headset output (left)
6	AGND	Analog ground
7	TESTEN	Test enable input HIGH enables the UART signal.
8	HOOK_SW-	Hook switch Input This is LOW while the switch is being pressed.
9	BTTXD	BTUART transmission
10	BTRXD	BTUART reception

5-25 Interface between Circuit Boards

The connector specifications for connecting the main circuit board and key circuit board are given below.

Model No.:	Main circuit board	30P3.5-JMCS-G-TF	plug	JST
	Key circuit board	30R-JMCS-G-TF(S)	receptor	JST
Engagement height:	3.5 mm			

<JSW Connector>

PIN	NAME	GPIO	FUNCTION
1	3.3 V	-	3.3 V power
2	3.3 V	-	3.3 V power
3	3.3 V	-	3.3 V power
4	KP_MKIN0	100	Keypad matrix input 0
5	KP_MKIN1	101	Keypad matrix input 1
6	KP_MKIN2	102	Keypad matrix input 2
7	KP_MKIN3	97	Keypad matrix input 3
8	KP_MKOUT0	103	Keypad matrix output 0
9	KP_MKOUT1	104	Keypad matrix output 1
10	KP_MKOUT2	105	Keypad matrix output 2
11	KP_MKOUT3	106	Keypad matrix output 3
12	KP_MKOUT4	107	Keypad matrix output 4
13	KP_MKOUT5	108	Keypad matrix output 5
14	KP_MKOUT6	96	Keypad matrix output 6
15	GND	-	GROUND
16	AGND	-	Analog ground
17	INT_MIC	-	Built-in microphone input
18	AGND	-	Analog ground
19	GND	-	GROUND
20	POINT_RST-	97	Pointer sensor reset
21	POINT_INT-	9	Pointer sensor interrupt
22	KEY_LED	52	Lights up key lighting LED
23	POINT_CS	22	Pointer sensor chip select
24	SCLK	99	Pointer sensor serial clock
25	SDI	36	Pointer sensor data input
26	SDO	77	Pointer sensor data output
27	PWR_SW	0	Power key
28	-	-	-
29	GND	-	GROUND
30	GND	-	GROUND

6 GPIO List

A list of the GPIO definition used in this handset is given below.

GPIO	NAME	ALT FUNCTION	DIRECTION	DEFAULT OUTPUT
0	PWR_SW	0	INPUT	-
1	GPIO_RES-	0	INPUT	0
2	SYS_EN	-	OUTPUT	-
3	PWR_SCL	0	BIDIRECTION	-
4	PWR_SDA	0	BIDIRECTION	-
5	PWR_CAP0	-	-	-
6	PWR_CAP1	-	-	-
7	PWR_CAP2	-	-	-
8	PWR_CAP3	-	-	-
9	POINT_INT-	0	INPUT	-
10	CF_IRQ	0	INPUT	-
11	AC97_IRQ	0	INPUT	-
12	CIF_DD7	2	INPUT	-
13	CF_BVD1	0	INPUT	-
14	WLAN_IRQ	0	INPUT	-
15	CS1-	2	OUTPUT	1
16	PWM0	2	OUTPUT	0
17	CIF_DD6	2	INPUT	-
18	HDSET_DET-	0	INPUT	-
19	CF_PWRON-	0	OUTPUT	1
20	VPEN	0	OUTPUT	1
21	ONEWIRE	0	BIDIRECTION	1
22	POINT_CS	0	OUTPUT	0
23	CIF_MCLK	1	OUTPUT	0
24	CIF_FV	1	INPUT	-
25	CIF_LV	1	INPUT	-
26	CIF_PCLK	2	INPUT	-
27	CIF_DD0	3	INPUT	-
28	AC_BITCLK	1	INPUT	-
29	AC_SDATAIN0	1	INPUT	-
30	AC_SDATAOUT	2	OUTPUT	0
31	AC_SYNC	2	OUTPUT	0

GPIO	NAME	ALT FUNCTION	DIRECTION	
32	MMCLK	2	OUTPUT	0
33	CS5-	2	OUTPUT	1
34	CF_CD1-	0	INPUT	-
35	CF_CD2-	0	INPUT	-
36	SDI	0	OUTPUT	0
37	CF_BVD2	0	INPUT	-
38	WLAN_STBY-	0	OUTPUT	0
39	WLAN_RESET-	0	OUTPUT	0
40	CAMERA_RESET-	0	OUTPUT	0
41	CAMERA_ON	0	OUTPUT	0
42	BTRXD	1	INPUT	-
43	BTTXD	2	OUTPUT	0
44	CAM1_EN	0	OUTPUT	0
45	CAM2_EN	0	OUTPUT	0
46	STD_RXD	2	INPUT	-
47	STD_TXD	1	OUTPUT	0
48	POE-	2	OUTPUT	1
49	PWE-	2	OUTPUT	1
50	PIOR-	2	OUTPUT	1
51	PIOW-	2	OUTPUT	1
52	KEY_LED	0	OUTPUT	0
53	HOOK_SW-	0	INPUT	-
54	PCE2-	2	OUTPUT	1
55	PREG-	2	OUTPUT	1
56	PWAIT-	1	INPUT	-
57	IOIS16-	1	INPUT	-
58	L_DD0	2	OUTPUT	0
59	L_DD1	2	OUTPUT	0
60	L_DD2	2	OUTPUT	0
61	L_DD3	2	OUTPUT	0
62	L_DD4	2	OUTPUT	0
63	L_DD5	2	OUTPUT	0

GPIO	NAME	ALT FUNCTION	DIRECTION	
64	L_DD6	2	OUTPUT	0
65	L_DD7	2	OUTPUT	0
66	L_DD8	2	OUTPUT	0
67	L_DD9	2	OUTPUT	0
68	L_DD10	2	OUTPUT	0
69	L_DD11	2	OUTPUT	0
70	L_DD12	2	OUTPUT	0
71	L_DD13	2	OUTPUT	0
72	L_DD14	2	OUTPUT	0
73	L_DD15	2	OUTPUT	0
74	L_FCLK	2	OUTPUT	0
75	L_LCLK	2	OUTPUT	0
76	L_PCLK	2	OUTPUT	0
77	SDO	0	INPUT	-
78	USB_PULLUP	0	OUTPUT	0
79	PSKTSEL	1	OUTPUT	1
80	IRDA_SD	0	OUTPUT	0
81	CHG_EN-	0	OUTPUT	0
82	SD_PWRON-	0	OUTPUT	1
83	USB_DETECT	0	INPUT	-
84	PMIC_SLEEP	0	OUTPUT	1
85	PCE1-	1	OUTPUT	1
86	CF_RESET-	0	OUTPUT	0
87	POINT_RST-	0	OUTPUT	0
88	SPKR_EN	0	OUTPUT	0
89	RCV_EN	0	OUTPUT	0
90	CIF_DD4	3	INPUT	-
91	CIF_DD5	3	INPUT	-
92	MMDAT0	1	BIDIRECTION	-
93	SD_DETECT	0	INPUT	-
94	VIBRATOR_ON	0	OUTPUT	0
95	LCD_CS	0	OUTPUT	0

GPIO	NAME	ALT FUNCTION	DIRECTION	
96	KP_MKOUT6	3	OUTPUT	1
97	KP_MKIN3	3	INPUT	-
98	LCD_RESET-	0	OUTPUT	0
99	SCLK	0	OUTPUT	0
100	KP_MKIN0	1	INPUT	-
101	KP_MKIN1	1	INPUT	-
102	KP_MKIN2	1	INPUT	-
103	KP_MKOUT0	2	OUTPUT	1
104	KP_MKOUT1	2	OUTPUT	1
105	KP_MKOUT2	2	OUTPUT	1
106	KP_MKOUT3	2	OUTPUT	1
107	KP_MKOUT4	2	OUTPUT	1
108	KP_MKOUT5	2	OUTPUT	1
109	MMDAT1	1	BIDIRECTION	-
110	MMDAT2	1	BIDIRECTION	-
111	MMDAT3	1	BIDIRECTION	-
112	MMCMD	1	BIDIRECTION	-
113	AC_RES-	2	OUTPUT	0
114	CIF_DD1	1	INPUT	-
115	CIF_DD3	2	INPUT	-
116	CIF_DD2	1	INPUT	-
117	I2C_SCL	1	BIDIRECTION	-
118	I2C_SDA	1	BIDIRECTION	-

7 TP List

TP	Signal Name	Comments	TP	Signal Name	Comments
TP1	BATT_FLT-		TP33	HDSET_DET-	
TP2	VDD_FLT-		TP34	R_OUT	
TP3	RESET_OUT-		TP35	HOOK_SW-	
TP4	XTAL1	13 MHz	TP36	SPKR_EN	
TP5	XTAL2	32.768 KHz	TP37	RCV_EN	
TP6	USB 5V	5.0 V	TP38	SPP	Speaker
TP7	USB GND	0 V	TP39	SPN	Speaker
TP8	VCC_USB	5.0 V	TP40	SPP	Receiver
TP9	USB_DETECT		TP41	SPN	Receiver
TP10	CAMERA_ON		TP42	VBAT	
TP11	VCC_CAM	2.5 V	TP43	BAT VCC	
TP12	WLAN VCC	3.15 V	TP44	BAT TS	
TP13	INT_MIC		TP45	BATT GND	
TP14	LCD_BL_ON		TP46	PWR_SW	
TP15	BACK LIGHT	12.0 V	TP47	GPIO_RESET-	
TP16	CF_CD2-		TP48	VCC_BACKUP	
TP17	CF_CD1-		TP49	VCC_3P3V	3.15 V
TP18	CF_PWRON-		TP50	VCC_1P8V	1.84 V
TP19	VCC_CF	3.15 V	TP51	VCC_CORE	1.30 V(Default)
TP20	SD_PWRON-		TP52	VCC_PLL	1.30 V
TP21	VCC_SD	3.15 V	TP53	VCC_RAM	1.10 V
TP22	SD_DETECT		TP54	RESET-	
TP23	VIB_ON-		TP55	VCC_BATT	3.0 V
TP24	VCC_VIB	3.15 V	TP56	CELL VCC	3.0 V
TP25	PWR_EN		TP57	MR-	
TP26	OSC1	24.576 MHz	TP58	TESTTXD	
TP27	MICOUT		TP59	TESTRXD	
TP28	MOUT2		TP60	VCC_MEM	1.80 V
TP29	MOUT+		TP61	VCC_SDRAM	1.80 V
TP30	MOUT-		TP62	KP_MKIN0	
TP31	EXT_MIC		TP63	KP_MKOUT4	
TP32	L_OUT				

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